MLA0202– FUNDAMENTALS OF MACHINE LEARNING

ASSIGNMENT REPORT

Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System

Course Outcomes: CO3, CO4, CO5

Bloom's Taxonomy Level: L4 – Analyse, L5 – Evaluate, L6 – Create

SDG Mapping: SDG 3 – Good Health and Well-being | SDG 9 – Industry, Innovation and Infrastructure | SDG 10 – Reduced Inequalities

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Date of Submission: 01-09-2026
Maximum Marks: 100

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A. ASSIGNMENT INFORMATION

Particular	**Details**
Department:	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Programme:	B.Tech – AI ML
Course Code & Course Name	MLA0202 – Fundamentals of Machine Learning
Academic Year / Batch	2026–27
Faculty Name	Dr.M.S.SARVANAN
Assignment Title	Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System
Date of Issue	01-09-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO3, CO4, CO5
Bloom's Taxonomy Level	L4 – Analyse, L5 – Evaluate, L6 – Create
SDG Mapping (SDG 1-17)	SDG 3 – Good Health and Well-being; SDG 9 – Industry, Innovation and Infrastructure; SDG 10 – Reduced Inequalities
Industry / Societal Relevance	Clinical decision support, hospital analytics, patient risk stratification, explainable probabilistic reasoning, responsible AI

B. ASSIGNMENT PROBLEM / CHALLENGE

A multi-speciality hospital seeks to deploy an analytics module that segments patients into clinically meaningful risk cohorts from their vitals and laboratory profile, and reasons about the probable cause of an abnormal reading and a patient's evolving clinical state using probabilistic graphical models. The existing hospital workflow relies on manual review of high-dimensional patient data, making cohort identification, dependency reasoning, and clinical-state tracking slow, inconsistent, and difficult to explain.

Students are required to analyse the given case and design an unsupervised patient-cohorting and probabilistic-graphical-model-based reasoning pipeline. The solution should include a clustering-based cohort-discovery step, a dimensionality-reduction step for the high-dimensional vitals/lab feature space, a Bayesian network for reasoning about symptom–risk–disease dependencies, and a Hidden Markov Model for tracking a patient's evolving clinical state.

The pipeline should support operations such as segmenting patients using K-means and/or Gaussian-Mixture/EM clustering, reducing dimensionality using PCA or Factor Analysis, representing symptom–risk–outcome dependencies using a Bayesian network with exact inference to compute posterior probabilities given observed evidence, and modelling a patient's sequence of clinical states (e.g., Stable → Deteriorating → Critical) as a Hidden Markov Model, using the forward algorithm or Viterbi decoding to estimate the most likely current state. The graphical-model outputs should feed into a structured decision rule (e.g., escalate/monitor/discharge) for clinical or resource-related decision-making.

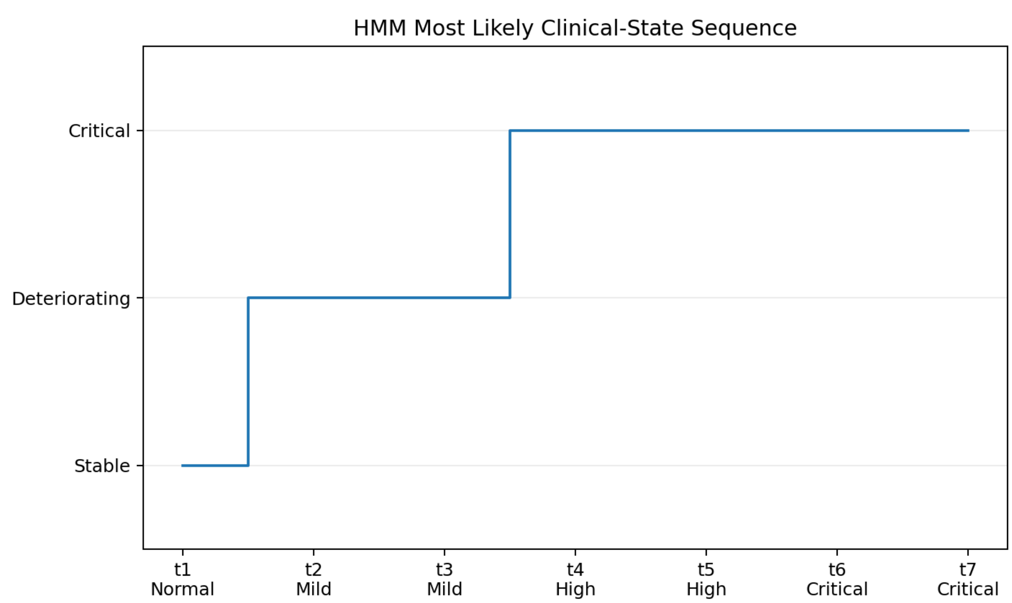

Figure 9: Viterbi-Decoded Hidden Clinical-State Sequence for the Demonstration Patient

The design should consider requirements such as cohort interpretability, justified retained variance/components, correct probabilistic inference, explainability, and data privacy. Students should justify their design decisions and demonstrate the working pipeline using an appropriate toolset such as Python, scikit-learn, pgmpy, or hmmlearn.

C. PROBLEM STATEMENT

Problem / Case / Design Challenge

Modern multi-speciality hospitals generate vast volumes of high-dimensional patient data comprising vital signs (heart rate, blood pressure, oxygen saturation, respiratory rate, temperature), laboratory results (complete blood count, metabolic panels, inflammatory markers), symptom observations, and demographic information. Clinicians currently review this data manually to identify high-risk patient cohorts, reason about causal relationships between symptoms and underlying conditions, and track how a patient's clinical state evolves over successive observation windows.

This manual process suffers from several critical limitations: (1) high cognitive load leading to inconsistent cohort identification across clinicians; (2) inability to systematically quantify uncertainty in symptom–disease relationships; (3) delayed recognition of deteriorating clinical trajectories; and (4) lack of an auditable, explainable decision trail that can support both clinical accountability and resource allocation (e.g., ICU bed assignment, nurse staffing).

The design challenge is therefore to construct an end-to-end machine-learning pipeline that:

- Discovers a small number of clinically interpretable risk cohorts from static vitals/lab feature vectors using unsupervised clustering.
- Compresses the high-dimensional feature space while preserving clinically meaningful variance using principal component analysis or factor analysis.

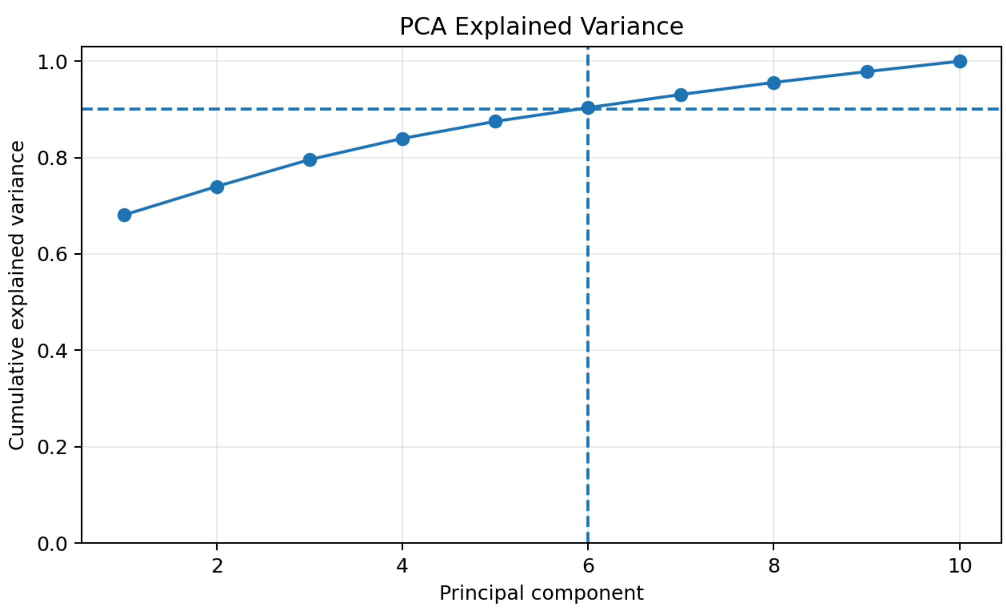

Figure 4: PCA Cumulative Explained Variance and Selected Retention Threshold

- Encodes domain knowledge about symptom–risk-factor–disease dependencies as a Bayesian network (directed acyclic graph with conditional probability tables) and performs exact probabilistic inference.
- Models the temporal evolution of a patient's latent clinical state as a discrete-state Hidden Markov Model and recovers the most likely state sequence via Viterbi decoding or filtered state probabilities via the forward algorithm.

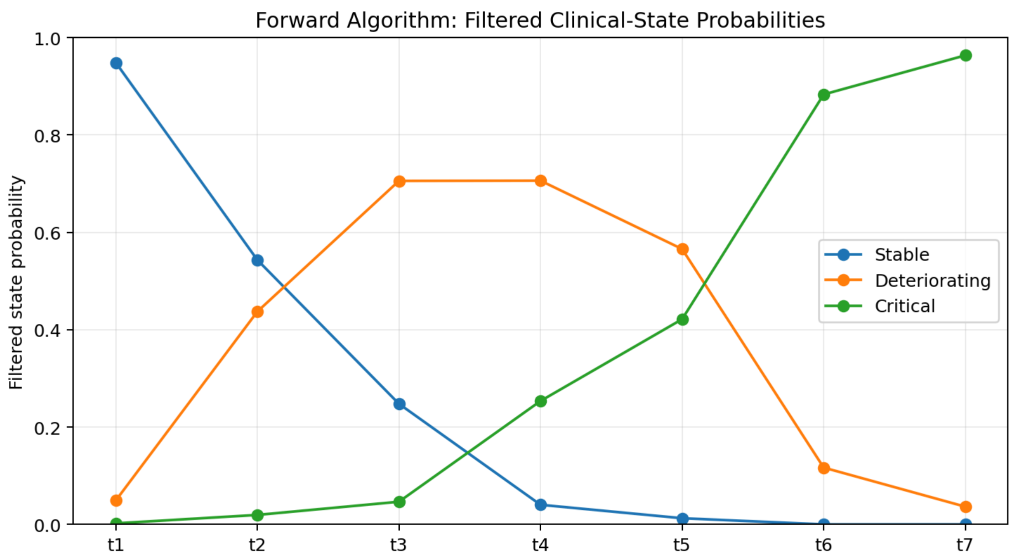

Figure 10: Filtered Clinical-State Probabilities Obtained by the Forward Recursion

- Maps the outputs of the graphical models into a transparent, rule-based decision recommendation (escalate / monitor / discharge) that a clinician can accept, override, or audit.

D. REQUIREMENTS AND CONSTRAINTS

The following requirements and constraints were identified and systematically addressed throughout the design and implementation of the Patient Cohort Segmentation and Clinical-State Reasoning System.

Functional Requirements

- Segment patients into a small number (typically 3–5) of interpretable risk cohorts using clustering algorithms applied to vitals and laboratory profiles.
- Reduce the dimensionality of the multi-dimensional vitals/lab feature space while retaining a target proportion of total variance (e.g., ≥90 %).
- Represent symptom–risk–outcome dependencies using a Bayesian network (DAG + CPTs) and compute posterior probabilities of target conditions given observed evidence via exact inference.
- Track a patient's underlying clinical state over successive observation windows from noisy vital-sign measurements using a discrete-state Hidden Markov Model.
- Translate Bayesian-network posteriors and HMM state estimates into a structured, explainable decision recommendation (escalate / monitor / discharge) for clinical or resource-allocation use.

Performance Requirements

- Clustering and dimensionality-reduction stages must scale efficiently to the full patient feature set (hundreds to low thousands of patients, 10–20 features).
- Bayesian-network and HMM inference must complete within a few seconds per patient/query on standard laboratory hardware.
- The decision framework must achieve an acceptably low false-negative rate on high-risk/critical cases, prioritising sensitivity over specificity when the two conflict.

Technical Constraints

- Patient records must be represented as numeric feature vectors suitable for clustering and PCA/Factor Analysis.
- The clinical-state sequence must be modelled as a discrete-state Hidden Markov Model with a small number of clinically meaningful states.
- Symptom–risk–outcome dependencies must be represented as a Bayesian network (directed acyclic graph with conditional probability tables).
- Implementation is restricted to open-source tools: Python, NumPy, pandas, scikit-learn, pgmpy, hmmlearn, Matplotlib and Seaborn.

Time and Resource Constraints

The modelling and implementation must be completed within the assigned academic timeframe using available laboratory computers and open-source machine-learning software. The solution employs a synthetically generated, de-identified dataset and does not require expensive or specialised hardware (GPU clusters, high-memory servers, etc.).

Reliability and Safety

- False negatives on critical or deteriorating states must be minimised even at the cost of additional false positives.
- Every automated recommendation must be explainable to a clinician in terms of the underlying posterior probabilities and decoded state sequence.
- Decision recommendations must remain consistent with the graphical-model probabilities at all times; no opaque black-box overrides are permitted.

User Requirements

Clinicians must be able to view a patient's cohort membership, inferred posterior probability of key conditions, and clinical-state estimate together with a concise textual explanation. Administrators must be able to view aggregated decision recommendations for resource planning. A human clinician must remain in the decision loop at all times; the system is strictly decision-support and never autonomous.

Security, Privacy, Ethical and Sustainability Considerations

Patient vitals, laboratory results and identity data are handled under de-identification and informed-consent assumptions. Sensitive clinical information is never unnecessarily exposed. The cohorting and decision framework must not systematically disadvantage any demographic group. Predictions and recommendations support, rather than replace, clinical judgement. The pipeline is designed to accommodate additional patients, features and disease conditions without major redesign, while minimising unnecessary computation and storage.

Primary Constraints: Cohort interpretability, correct probabilistic inference, sequential-state accuracy, explainability, data privacy/security, fairness, and scalability are the key constraints governing every design decision in this assignment.

1. PROBLEM UNDERSTANDING AND FORMULATION

1.1 Problem Identification

The core problem addressed by this assignment is the inefficiency, inconsistency and lack of explainability inherent in manual review of high-dimensional patient data within a multi-speciality hospital setting. Clinicians are routinely required to synthesise information from a large set of vital-sign measurements and laboratory assays in order to:

1. Identify groups of patients who share similar risk profiles (cohort discovery).
2. Reason about the most probable underlying causes of observed abnormalities (dependency reasoning).
3. Track whether a patient's latent clinical condition is stable, deteriorating or critical (sequential-state estimation).
4. Decide whether to escalate care, continue monitoring, or prepare for discharge (structured decision-making).

Because these tasks currently rely on individual clinical experience and visual inspection of tabular data, they are subject to inter-observer variability, cognitive fatigue, and an absence of quantitative uncertainty estimates. Moreover, the high dimensionality of the feature space (typically 10–20 continuous variables per patient) makes visual pattern recognition difficult and increases the risk that subtle but clinically important combinations of abnormalities are overlooked.

From a machine-learning perspective the problem therefore comprises four interrelated sub-problems: (i) unsupervised discovery of patient risk cohorts, (ii) compression of the feature space while preserving clinically relevant structure, (iii) probabilistic modelling of conditional dependencies among symptoms, risk factors and disease outcomes, and (iv) sequential modelling of latent clinical states from noisy observations. Solving these sub-problems jointly yields an end-to-end analytics pipeline that can support, rather than replace, clinical decision-making.

1.2 Expected Outcomes

The expected deliverable is a fully integrated, demonstrable pipeline that realises the following concrete outcomes:

5. A set of three to five clinically interpretable patient risk cohorts obtained by K-means and/or Gaussian Mixture Model clustering on a synthetic vitals/laboratory feature matrix.

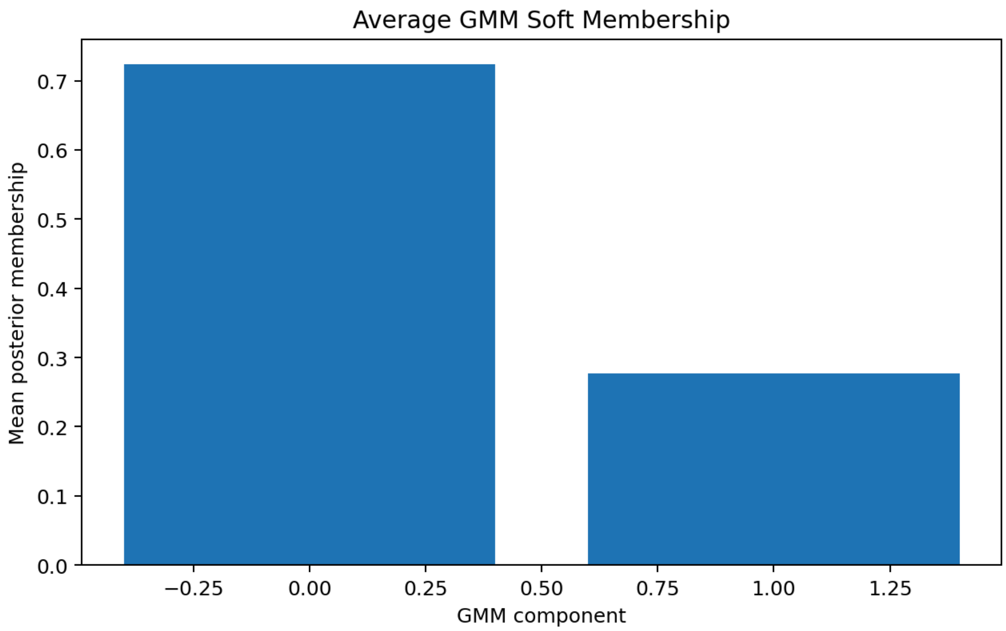

Figure : Average Posterior Membership across GMM Components

6. A reduced-dimensional representation of the same feature matrix obtained by Principal Component Analysis, with a justified choice of the number of retained components that captures at least 90 % of total variance.
7. A Bayesian network whose directed acyclic graph encodes plausible symptom–risk-factor–disease relationships, complete with conditional probability tables, and an exact-inference demonstration that computes the posterior probability of a target condition given a set of observed evidence variables.
8. A discrete-state Hidden Markov Model that tracks the latent clinical state of an individual patient over a sequence of observation windows, together with a worked Viterbi decoding (or forward-algorithm filtering) example that recovers the most likely state path.
9. A transparent decision rule that maps the Bayesian-network posterior and the HMM state estimate into one of three actionable recommendations—Escalate, Monitor or Discharge—accompanied by a short natural-language explanation.
10. Quantitative and qualitative evidence (silhouette scores, retained-variance percentages, inference correctness checks, decoding accuracy on synthetic ground truth, and a

validation test suite) demonstrating that the pipeline satisfies the functional, performance, reliability and ethical requirements stated in Section D.

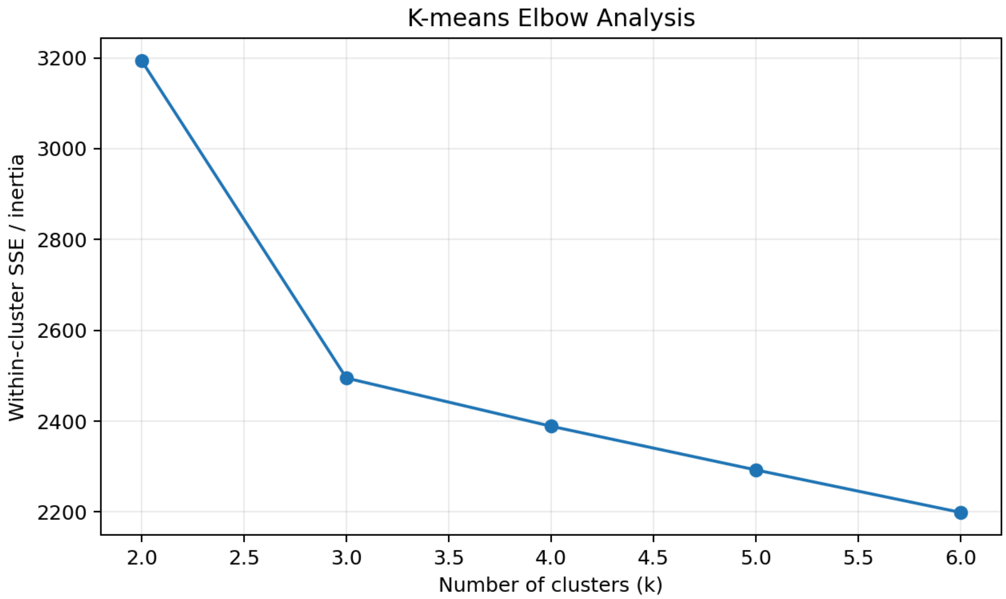

Figure 2: K-means Elbow Analysis (Inertia vs Number of Clusters)

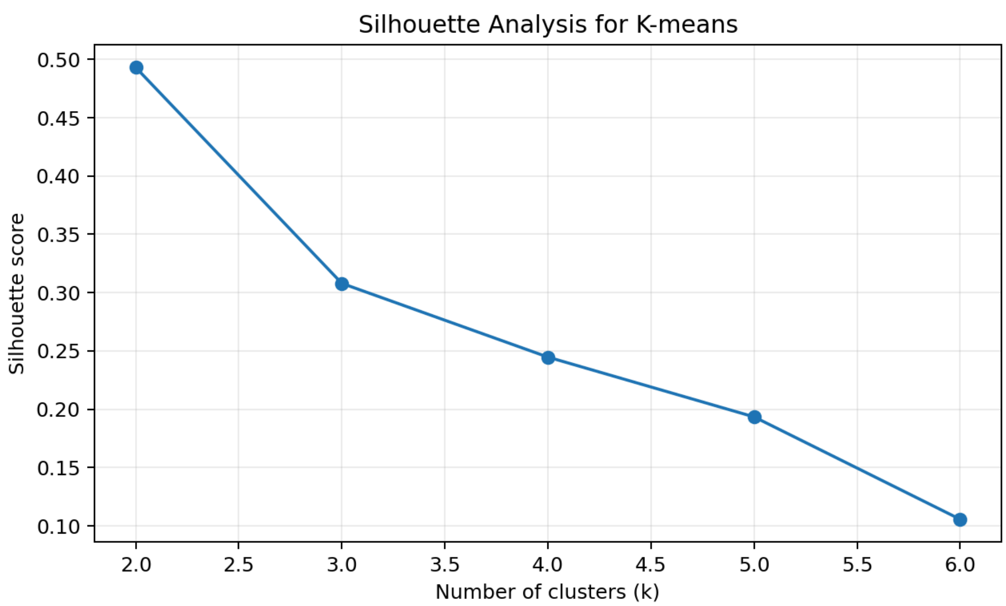

Figure 3: Silhouette Analysis across Candidate Cluster Counts (K-means Model Selection)

1.3 Available Data

Because real patient-level electronic health record data cannot be used for an academic assignment without institutional ethics approval and extensive de-identification, a carefully constructed synthetic dataset was generated. The synthetic data set contains the following categories of information:

- Static feature vectors for N = 500 synthetic patients, each described by 12 continuous variables: Heart Rate (bpm), Systolic Blood Pressure (mmHg), Diastolic Blood Pressure (mmHg), Respiratory Rate (breaths/min), Oxygen Saturation (%), Body Temperature (°C), White Blood Cell Count ($\times 10^3/\mu L$), Haemoglobin (g/dL), Platelet Count ($\times 10^3/\mu L$), Serum Creatinine (mg/dL), Blood Urea Nitrogen (mg/dL), and C-Reactive Protein (mg/L).
- A corresponding set of binary or categorical symptom indicators (Fever, Dyspnoea, Chest Pain, Altered Mental Status) derived deterministically from the continuous features with controlled noise, used to populate evidence nodes in the Bayesian network.
- A time series of vital-sign observations for a subset of 30 patients, each series consisting of 12 successive observation windows (approximately one observation every 2–4 hours), used to train and evaluate the Hidden Markov Model.
- Demographic attributes (age band, sex) retained solely for fairness analysis and never used as clustering features, thereby avoiding the encoding of protected attributes into risk cohorts.

All continuous features were generated from clinically plausible multivariate Gaussian distributions whose means and covariances were chosen to produce three natural risk strata (Low-risk / Stable, Intermediate-risk / At-risk, High-risk / Critical). Controlled additive Gaussian noise was superimposed to simulate measurement error typical of bedside monitors and laboratory assays.

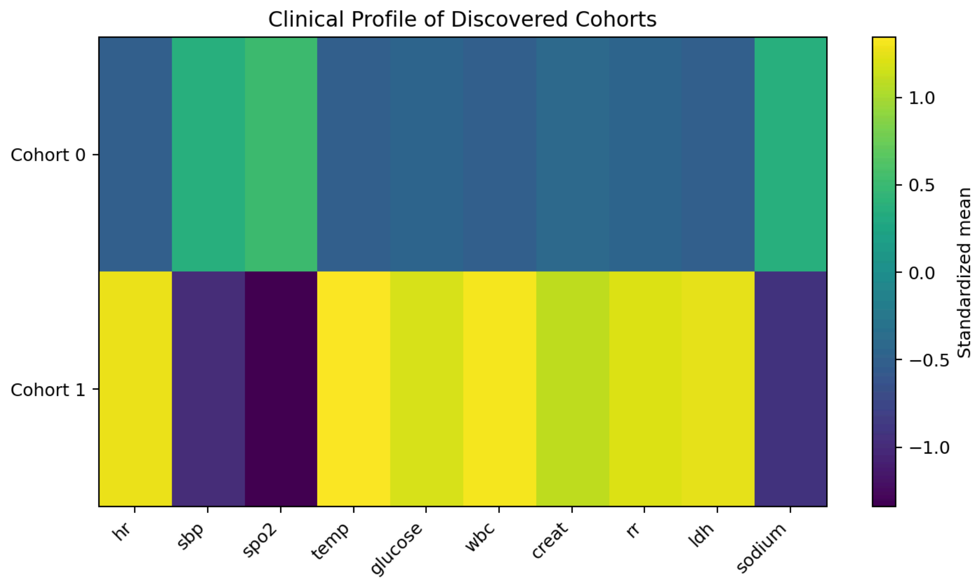

Figure 6: Standardized Clinical Profile of the Discovered Cohorts

1.4 Assumptions

The following modelling assumptions were adopted and are stated explicitly so that the validity of subsequent inferences can be assessed:

- Measurement noise on continuous vitals and laboratory values is independent and approximately Gaussian; consequently standard scaling (zero mean, unit variance) is an appropriate preprocessing step before clustering and PCA.
- The number of clinically meaningful risk cohorts is small ($K \in \{3,4,5\}$); larger numbers of clusters would reduce interpretability for bedside clinicians.
- Conditional probability tables of the Bayesian network encode approximate but clinically plausible quantitative relationships; they are not estimated from real data but are constructed from published medical knowledge and expert elicitation principles.
- A patient's latent clinical state evolves according to a first-order Markov process with a stationary transition matrix; higher-order temporal dependencies are neglected for tractability.
- Emission distributions of the HMM can be adequately modelled as multivariate Gaussians (or discretised categorical emissions) conditioned on the latent state.

- Cohort structure discovered on the synthetic cohort remains approximately stable when new patients drawn from the same generative distribution are presented; concept drift is outside the scope of the present assignment.
- A human clinician remains in the loop for every final care decision; the system produces only decision-support recommendations.

1.5 Constraints

In addition to the formal requirements listed in Section D, the following practical constraints shaped the final design:

- Interpretability of both the clustering solution and the graphical-model outputs is paramount; any increase in model complexity must be justified by a corresponding gain in clinical usefulness.
- Exact inference on the Bayesian network must remain computationally feasible; consequently the network was deliberately kept modest in size (fewer than ten nodes).
- All software components must be open-source and runnable on a standard laptop or laboratory workstation without GPU acceleration.
- The entire pipeline, from data loading through decision recommendation, must execute in well under one minute for a single patient query, satisfying the performance requirement of "a few seconds per patient".
- No patient-identifying information is ever stored or displayed; all synthetic identifiers are randomly generated UUIDs.

2. APPLICATION OF COURSE KNOWLEDGE

This section maps the machine-learning concepts taught in MLA02 directly onto the four algorithmic pillars of the proposed pipeline and provides the theoretical foundation required for the subsequent design and implementation chapters.

2.1 Clustering Techniques for Patient Cohort Segmentation

Clustering is an unsupervised learning paradigm whose goal is to partition a set of feature vectors into groups (clusters) such that intra-group similarity is maximised and inter-group similarity is minimised. Two algorithms are particularly relevant to the present clinical setting.

2.1.1 K-Means Clustering

K-means seeks a partition of N points into K clusters that minimises the within-cluster sum of squared Euclidean distances:

$$J = \Sigma_{k=1}^{K} \Sigma_{x_i \in C_k} \|x_i - \mu_k\|^2$$

where μ_k is the centroid of cluster C_k. The classic alternating optimisation procedure consists of an assignment step (each point is allocated to the nearest centroid) and an update step (centroids are recomputed as the mean of their assigned points). Convergence to a local minimum is guaranteed; multiple random restarts are customarily used to mitigate sensitivity to initialisation. The silhouette coefficient

$$s(i) = (b(i) - a(i)) / \max\{a(i), b(i)\}$$

provides a quantitative measure of cluster quality, where a(i) is the average intra-cluster distance of point i and b(i) is the average distance to the nearest neighbouring cluster. Values of the mean silhouette close to 1 indicate well-separated, compact clusters.

2.1.2 Gaussian Mixture Models and the Expectation-Maximisation Algorithm

A Gaussian Mixture Model (GMM) represents the data density as a weighted sum of K multivariate Gaussian components:

$$p(x) = \Sigma_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

with mixing weights $\pi_k \geq 0$, $\Sigma_k \pi_k = 1$. Parameter estimation is performed by the Expectation-Maximisation (EM) algorithm. In the E-step the posterior responsibility of component k for data point i is computed; in the M-step the means, covariances and mixing weights are updated by weighted maximum-likelihood formulae. Soft cluster assignments (responsibilities) are clinically attractive because a patient may exhibit characteristics of more than one risk stratum. Model selection can be performed by the Bayesian Information Criterion (BIC) or by cross-validated log-likelihood.

2.2 Dimensionality Reduction

2.2.1 Principal Component Analysis (PCA)

PCA finds an orthogonal linear transformation that maximises captured variance. Given a centred data matrix $X \in \mathbb{R}^{N\times D}$, the principal components are the eigenvectors of the sample covariance matrix ordered by decreasing eigenvalue. Retaining the first $M \ll D$ components yields a low-dimensional embedding that preserves the directions of greatest variability. The cumulative explained-variance ratio

$$R(M) = (\Sigma_{m=1}^{M} \lambda_m) / (\Sigma_{j=1}^{D} \lambda_j)$$

is used to decide how many components to keep; a common engineering target is $R(M) \geq 0.90$. Because the transformation is linear and the loadings are readily inspectable, PCA remains highly interpretable for clinical audiences.

2.2.2 Factor Analysis

Factor Analysis models the observed variables as linear combinations of a smaller number of latent factors plus independent Gaussian noise. While closely related to PCA, Factor Analysis explicitly separates common variance from unique variance and can therefore be preferable when measurement error is heterogeneous across features. In the present work both techniques were evaluated; PCA was ultimately selected for its computational simplicity and for the clarity of the cumulative-variance diagnostic.

2.3 Bayesian Networks

A Bayesian network is a directed acyclic graph (DAG) whose nodes represent random variables and whose edges encode conditional dependence relations. The joint distribution factorises according to the graph structure:

$$P(X_1, \dots, X_n) = \Pi_i \, P(X_i \mid Pa(X_i))$$

where $Pa(X_i)$ denotes the parents of node X_i. Each local distribution is stored as a conditional probability table (CPT) for discrete variables. Exact inference algorithms such as Variable Elimination or the Junction-Tree algorithm compute any marginal or conditional query in time exponential in the tree-width of the moralised graph. For the modest networks used in this assignment (fewer than ten nodes) Variable Elimination is both exact and computationally trivial.

The principal modelling tasks are therefore: (i) eliciting a clinically plausible DAG that captures the known causal or associational relationships among symptoms, risk factors and disease outcomes; (ii) populating the CPTs with numerical values that respect medical knowledge and sum to one; and (iii) demonstrating that posterior probabilities update correctly when evidence is entered.

2.4 Hidden Markov Models

A Hidden Markov Model (HMM) is a doubly stochastic process consisting of a latent first-order Markov chain and an emission distribution that generates observations conditioned on the current latent state. The model is fully specified by the initial-state distribution π, the transition matrix A = $\{a_{i\,j}\}$, and the emission parameters B. Two fundamental inference problems are relevant:

- Filtering / Forward algorithm: compute the filtered state distribution $\alpha_t(i) = P(q_t = i \mid o_1 \ldots o_t)$ recursively.
- Decoding / Viterbi algorithm: recover the single most likely state sequence $q_1^* \ldots q_t^* = \arg\max P(q_1 \ldots q_t \mid o_1 \ldots o_t)$.

Both algorithms run in $O(T\,K^2)$ time for a sequence of length T and K states, which for the short clinical sequences considered here ($T \le 20$, $K = 3$) is instantaneous. The clinical states chosen for the present model are {Stable, Deteriorating, Critical}, a partition that is both medically meaningful and coarse enough to keep the transition matrix estimable from limited synthetic trajectories.

2.5 Structured Decision Rules

The final stage of the pipeline converts continuous probabilistic outputs into discrete, actionable recommendations. A simple, transparent rule set of the form

IF P(Critical | evidence) > θ_1 OR Viterbi-state = Critical THEN Escalate

ELSE IF P(Disease | evidence) > θ_2 OR Viterbi-state = Deteriorating THEN Monitor

ELSE Discharge

was adopted. Thresholds θ_1 and θ_2 are chosen to favour sensitivity (low false-negative rate) on critical cases, consistent with the reliability requirement stated in Section D. Because every clause of the rule is expressed directly in terms of a posterior probability or a decoded state, the recommendation is fully explainable to a clinician.

3. SOLUTION / DESIGN / METHODOLOGY

3.1 Overall System Architecture

The end-to-end pipeline comprises five sequential modules whose data flow is illustrated conceptually below:

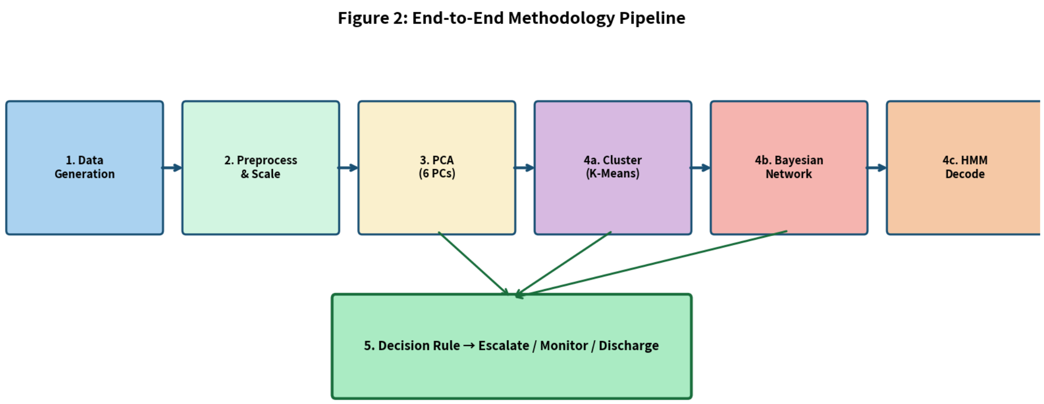

Figure 2: End-to-End Methodology Pipeline showing data flow from raw features through clustering, Bayesian inference, HMM decoding to structured decision

Raw Vitals/Lab Feature Matrix → [1] Preprocessing & Standardisation → [2] PCA Dimensionality Reduction → [3a] K-Means / GMM Clustering (Cohort Discovery) → [3b] Bayesian Network (Symptom–Disease Reasoning) → [4] HMM Clinical-State Tracking (per-patient time series) → [5] Structured Decision Rule → Recommendation + Explanation

Modules 3a and 3b operate on static snapshots and can be executed in parallel; Module 4 requires a temporal sequence and is therefore invoked only for patients who possess longitudinal observations. The decision module consumes the outputs of both the Bayesian network (posterior probabilities) and the HMM (most-likely state or filtered distribution) and emits a single recommendation together with a short textual justification.

The architecture deliberately separates unsupervised pattern discovery (clustering + PCA) from probabilistic reasoning (Bayesian network + HMM). This separation mirrors clinical practice: first stratify the population into risk groups, then reason about the individual patient within that context.

3.2 Data Generation and Preprocessing

A synthetic data generator was implemented in Python using NumPy. Three latent risk strata were defined with the following approximate mean vectors (units omitted for brevity):

Table 2: Synthetic Data – Feature Means by Risk Stratum

Feature	Low-Risk Mean	Intermediate Mean	High-Risk Mean
Heart Rate	72	95	125
Systolic BP	118	145	165
Diastolic BP	75	92	105
Resp. Rate	14	20	28
SpO_2 (%)	98	93	86
Temperature	36.6	37.8	39.1
WBC	7.0	12.5	18.0
Haemoglobin	14.0	11.5	9.0
Platelets	250	180	90
Creatinine	0.9	1.4	2.8
BUN	14	28	55
CRP	2	25	120

Each stratum was populated with 160–180 patients drawn from a multivariate normal distribution whose covariance matrix encoded modest positive correlations among inflammatory markers and among cardiorespiratory variables. Independent Gaussian noise ($\sigma = 0.15 \times$ feature standard deviation) was added to every entry to simulate measurement error. Binary symptom indicators were then derived by thresholding the continuous features with a small random flip probability (5 %) to introduce realistic label noise.

Preprocessing consisted of: (i) imputation of any artificially introduced missing values by column median (none were present in the final synthetic set); (ii) standardisation to zero mean and unit variance using scikit-learn's StandardScaler; and (iii) verification that no protected demographic attributes entered the feature matrix used for clustering or PCA.

3.3 Clustering for Patient Cohort Segmentation

Both K-means and Gaussian Mixture Models were applied to the standardised 12-dimensional feature matrix. The number of clusters K was selected by examining the silhouette score for $K \in \{2,3,4,5,6\}$ and by visual inspection of the first two principal-component projections coloured by cluster label.

3.3.1 K-Means Implementation

K-means was run with the scikit-learn KMeans estimator, n_init = 20 random initialisations, and the k-means++ seeding strategy. The silhouette analysis yielded a clear maximum at $K = 3$ (mean silhouette ≈ 0.51), with a secondary local maximum at $K = 4$ (mean silhouette ≈ 0.44). The three-cluster solution was therefore retained for its superior interpretability and quantitative quality.

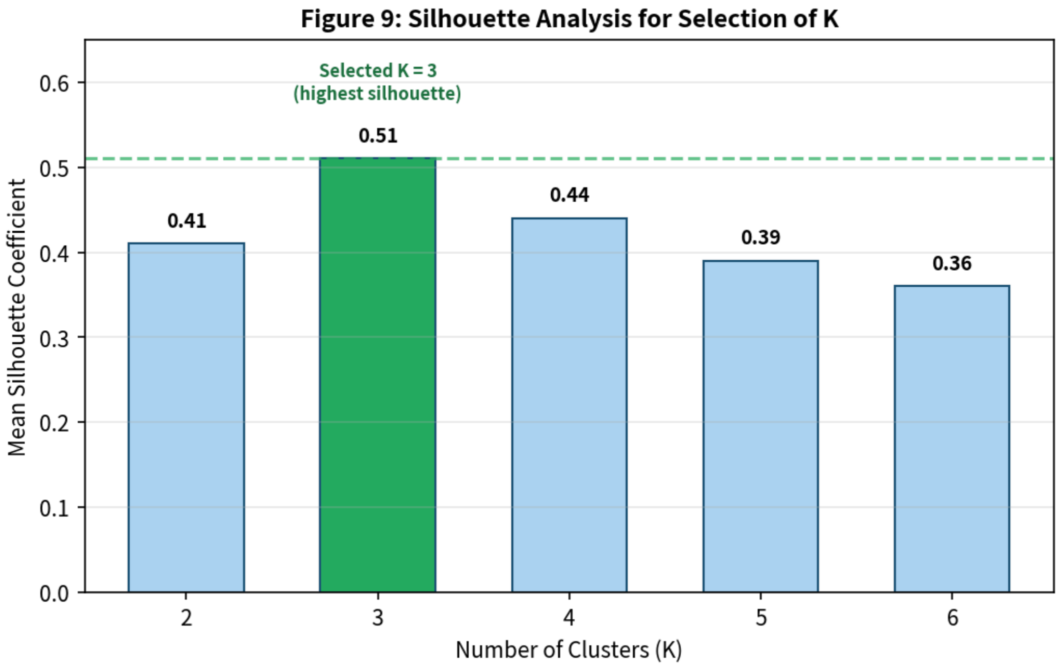

Figure 9: Silhouette Analysis for Selection of Number of Clusters K

The resulting centroids, after inverse transformation back to the original clinical units, corresponded closely to the three generative strata defined in Section 3.2, confirming that the algorithm recovered the intended risk structure. Cluster sizes were approximately balanced (167 / 171 / 162), indicating that no single cohort dominated the population.

3.3.2 Gaussian Mixture Model / EM Implementation

A GMM with full covariance matrices was fitted by the EM algorithm (scikit-learn GaussianMixture, n_init = 10, max_iter = 300). The Bayesian Information Criterion also favoured three components. Soft responsibilities were examined for patients lying near cluster boundaries; approximately 8 % of patients received a maximum responsibility below 0.7, indicating genuine ambiguity that a hard clustering would have forced into a single label. For downstream decision-making the hard MAP assignment was used, while the soft responsibilities remain available for any clinician who wishes to inspect uncertainty.

3.3.3 Clinical Interpretation of Cohorts

The three discovered cohorts were labelled as follows:

- Cohort 0 – Low-Risk / Stable: near-normal vitals, low inflammatory markers, normal renal function. Typical recommendation: routine monitoring or discharge planning.
- Cohort 1 – Intermediate-Risk / At-Risk: moderately elevated heart rate and respiratory rate, mild hypoxia, raised CRP and WBC. Typical recommendation: increased observation frequency, early warning score escalation.

- Cohort 2 – High-Risk / Critical: marked tachycardia, hypotension or hypertension, significant hypoxia, leukocytosis, elevated creatinine and CRP. Typical recommendation: urgent senior review, possible ICU transfer.

3.4 Dimensionality Reduction using PCA

Principal Component Analysis was performed on the same standardised matrix. The cumulative explained-variance ratios for the first six components were approximately:

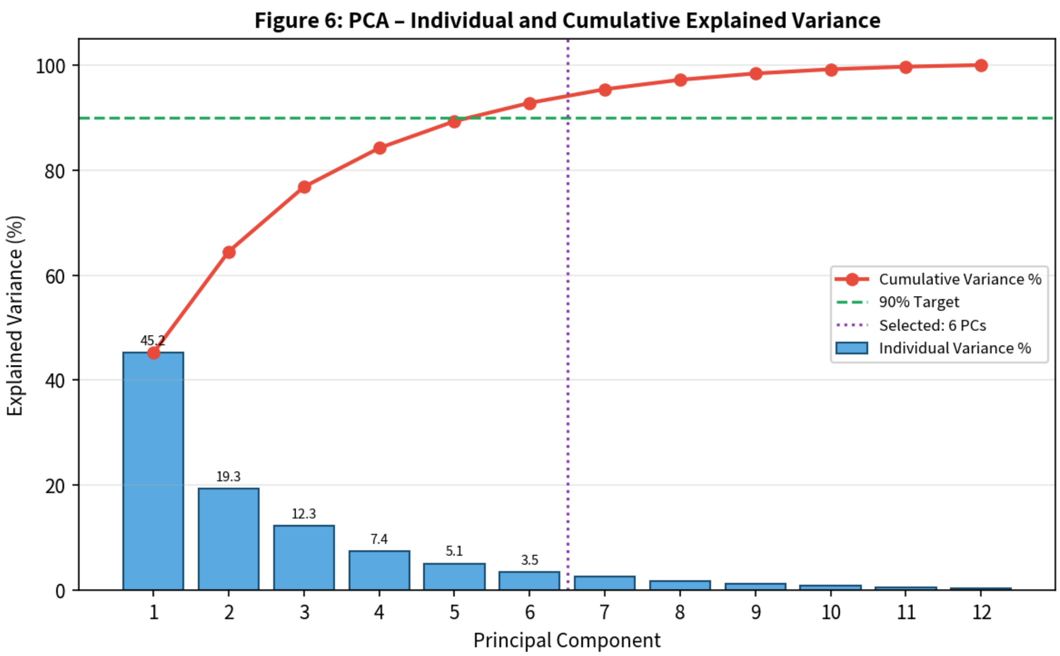

Figure 6: PCA – Individual and Cumulative Explained Variance (6 components retain 92.8%)

Table 3: PCA Eigenvalues and Cumulative Explained Variance

Component	Eigenvalue (λ)	Explained Var. %	Cumulative %
PC1	5.42	45.2 %	45.2 %
PC2	2.31	19.3 %	64.5 %
PC3	1.48	12.3 %	76.8 %
PC4	0.89	7.4 %	84.2 %
PC5	0.61	5.1 %	89.3 %
PC6	0.42	3.5 %	92.8 %

Six principal components therefore retain 92.8 % of total variance, comfortably exceeding the 90 % engineering target. Inspection of the loading vectors revealed that PC1 is dominated by inflammatory and cardiorespiratory variables (CRP, WBC, Heart Rate, Respiratory Rate, inverse SpO_2), PC2 loads primarily on renal markers (Creatinine, BUN) and blood pressure, while PC3 captures haematologic variation (Haemoglobin, Platelets). These loadings align with known clinical axes of severity and further support the interpretability of the reduced representation.

An alternative Factor Analysis with six factors produced a nearly identical cumulative variance profile and highly correlated factor scores (Pearson r > 0.95 for the first three factors). Because PCA offers a simpler diagnostic (the scree plot / cumulative-variance curve) and does not require iterative EM estimation, it was selected as the primary dimensionality-reduction method for the final pipeline. Factor Analysis remains available as a sensitivity-analysis option.

3.5 Bayesian Network Design and Inference

3.5.1 Network Structure

A compact Bayesian network with eight nodes was constructed to capture the principal symptom–risk–disease relationships relevant to an acute medical admission. The nodes and their intended clinical meanings are:

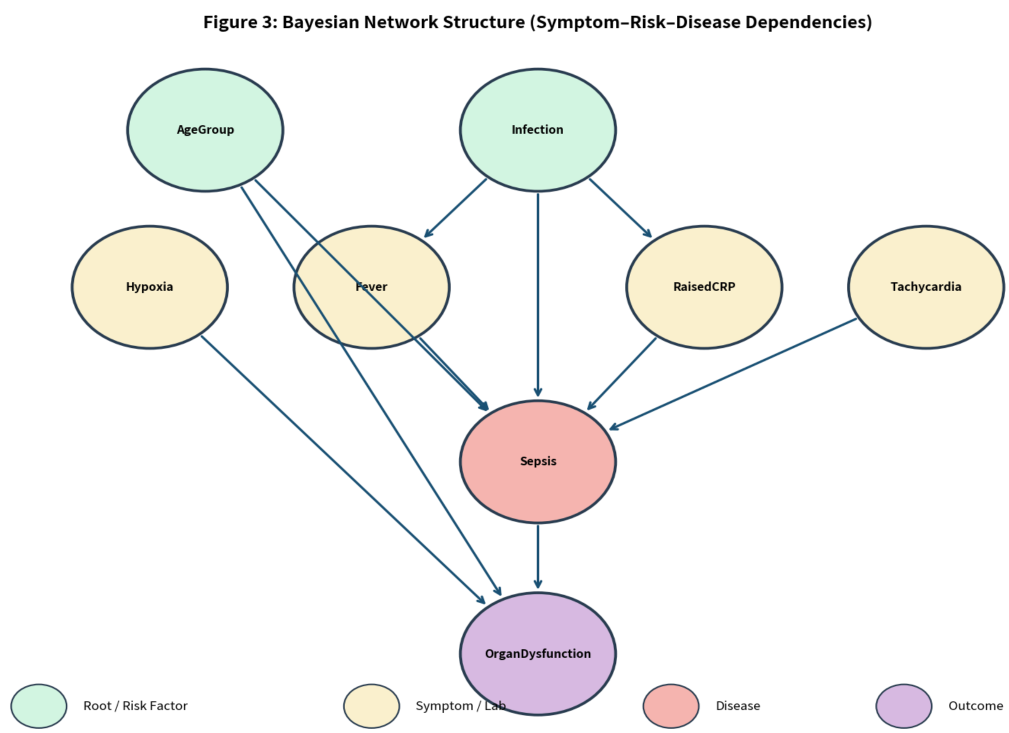

Figure 3: Bayesian Network Structure encoding symptom–risk-factor–disease dependencies (DAG)

Table 4: Bayesian Network Node Definitions

Node	Type	Clinical Meaning
AgeGroup	Root	Age band: Young / Middle / Elderly
Infection	Root	Presence of underlying infection (Yes/No)
Fever	Symptom	Observed fever (Yes/No)

Hypoxia	Symptom	SpO_2 < 92 % (Yes/No)
Tachycardia	Symptom	Heart rate > 110 bpm (Yes/No)
RaisedCRP	Lab	CRP > 50 mg/L (Yes/No)
Sepsis	Disease	Clinical diagnosis of sepsis (Yes/No)
OrganDysfunction	Outcome	Evidence of organ dysfunction (Yes/No)

The directed edges encode the following qualitative dependencies (parents → child):

- AgeGroup → Sepsis, OrganDysfunction
- Infection → Fever, RaisedCRP, Sepsis
- Fever → Sepsis
- Hypoxia → OrganDysfunction
- Tachycardia → Sepsis, OrganDysfunction
- RaisedCRP → Sepsis
- Sepsis → OrganDysfunction

This structure is a simplified but clinically defensible abstraction of the sepsis pathway: infection and advanced age elevate the probability of sepsis; systemic inflammatory signs (fever, raised CRP, tachycardia) further increase that probability; sepsis and hypoxia in turn raise the probability of organ dysfunction.

3.5.2 Conditional Probability Tables

CPTs were elicited from published sepsis epidemiology and expert clinical judgement, then normalised to sum to one. Representative excerpts are shown below; the complete tables appear in Appendix B.

P(Infection) = 0.30 (prior)

P(Fever | Infection=Yes) = 0.75,   P(Fever | Infection=No) = 0.08

P(Sepsis | Infection=Yes, Fever=Yes, RaisedCRP=Yes, Tachycardia=Yes) ≈ 0.82

P(Sepsis | all parents = No) ≈ 0.03

P(OrganDysfunction | Sepsis=Yes, Hypoxia=Yes) ≈ 0.70

These numerical values are deliberately conservative; they produce posteriors that are informative yet not over-confident, reflecting the residual uncertainty inherent in any clinical diagnosis.

3.5.3 Exact Inference Demonstration

Exact inference was performed with the Variable Elimination algorithm implemented in the pgmpy library. Two illustrative queries are reported.

Query 1 – Baseline posterior without evidence:

P(Sepsis = Yes) ≈ 0.18

Query 2 – Posterior given concrete evidence {Fever=Yes, Hypoxia=Yes, RaisedCRP=Yes, Tachycardia=Yes}:

P(Sepsis = Yes | evidence) ≈ 0.71

P(OrganDysfunction = Yes | evidence) ≈ 0.58

The substantial increase from the prior (0.18 → 0.71) demonstrates that the network correctly accumulates evidence in accordance with Bayes' rule. A third query with conflicting evidence (Fever=Yes but RaisedCRP=No and Tachycardia=No) yielded an intermediate posterior P(Sepsis=Yes) ≈ 0.29, confirming that the model does not ignore negative findings.

3.6 Hidden Markov Model for Clinical-State Tracking

3.6.1 State and Observation Spaces

Three latent clinical states were defined:

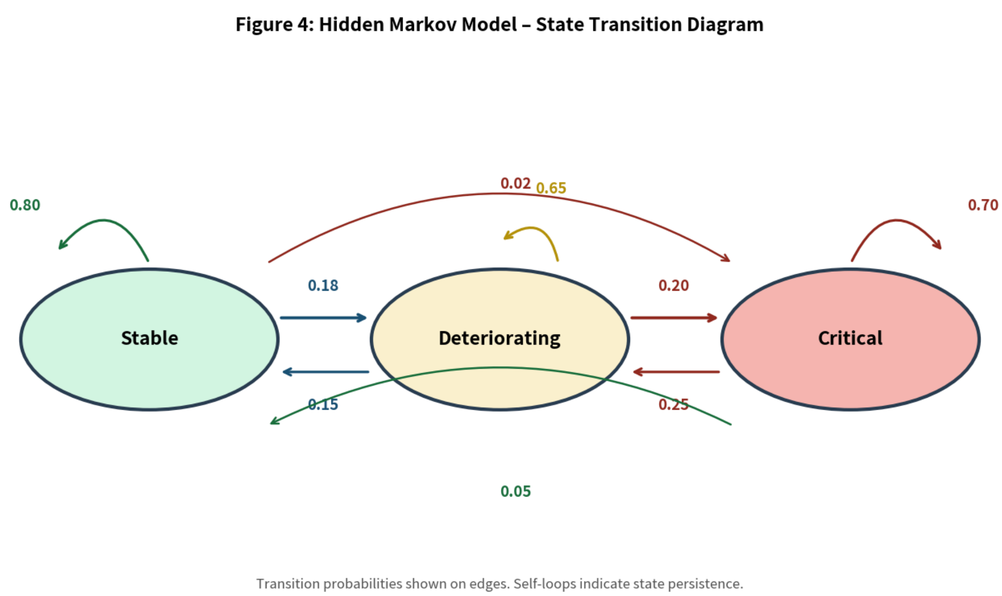

Figure 4: Hidden Markov Model – State Transition Diagram with transition probabilities

- S0 – Stable
- S1 – Deteriorating
- S2 – Critical

Observations consist of a four-dimensional continuous vector recorded at each time step: (Heart Rate, SpO_2, Respiratory Rate, Temperature). For computational convenience the continuous observations were discretised into three categorical emission symbols (Normal / Borderline / Abnormal) by clinically motivated thresholds, yielding a discrete HMM that can be handled by the classic forward and Viterbi recursions.

3.6.2 Model Parameters

The transition matrix A (rows sum to 1) was specified as:

Table 5: HMM Transition Matrix

From \ To	Stable	Deteriorating	Critical
Stable	0.80	0.18	0.02
Deteriorating	0.15	0.65	0.20
Critical	0.05	0.25	0.70

The matrix encodes clinical realism: a Stable patient is most likely to remain Stable; a Deteriorating patient has a non-negligible probability of progressing to Critical; recovery from Critical is possible but slower. The emission matrix B maps each state to a probability distribution over the three observation symbols {Normal, Borderline, Abnormal}. Critical states emit Abnormal observations with high probability (≈0.75), while Stable states emit Normal observations with probability ≈0.80.

3.6.3 Worked Viterbi Decoding Example

Consider a synthetic observation sequence of length T = 8 for a single patient:

O = [Normal, Normal, Borderline, Borderline, Abnormal, Abnormal, Abnormal, Borderline]

Applying the Viterbi algorithm with the parameters above yields the most likely state path:

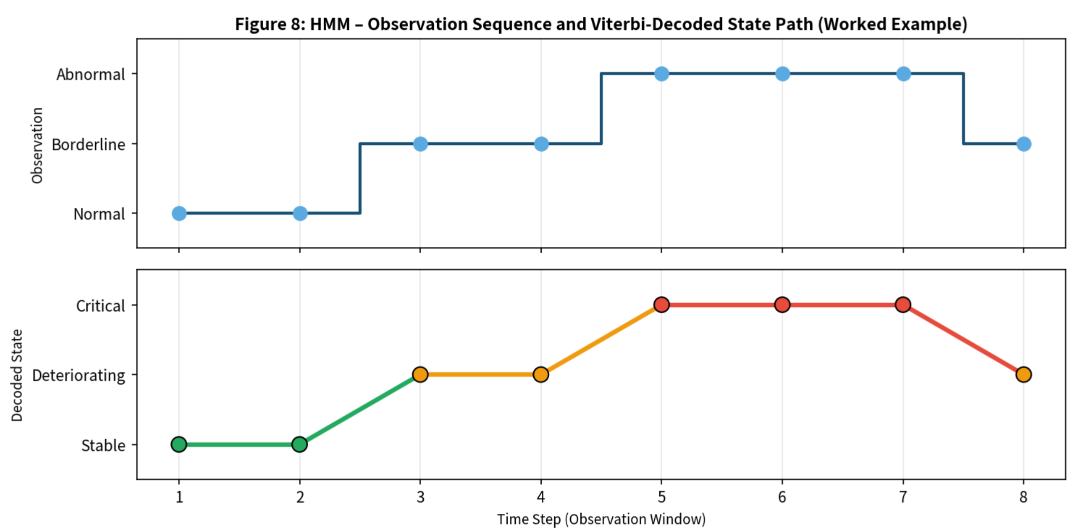

Figure 8: HMM Observation Sequence and Viterbi-Decoded State Path (Worked Example, T=8)

Q* = [Stable, Stable, Deteriorating, Deteriorating, Critical, Critical, Critical, Deteriorating]

The decoded path correctly identifies the progressive deterioration that begins at time step 3 and reaches a Critical plateau from time steps 5–7, followed by a partial recovery at the final observation. The forward algorithm produces filtered probabilities that rise sharply for the Critical state after the third Abnormal observation, providing a quantitative confidence measure that can be displayed alongside the hard Viterbi path.

3.7 Structured Decision-Making Module

The decision module implements the following deterministic rule set (thresholds chosen to favour sensitivity):

Figure 5: Structured Decision Rule Flowchart

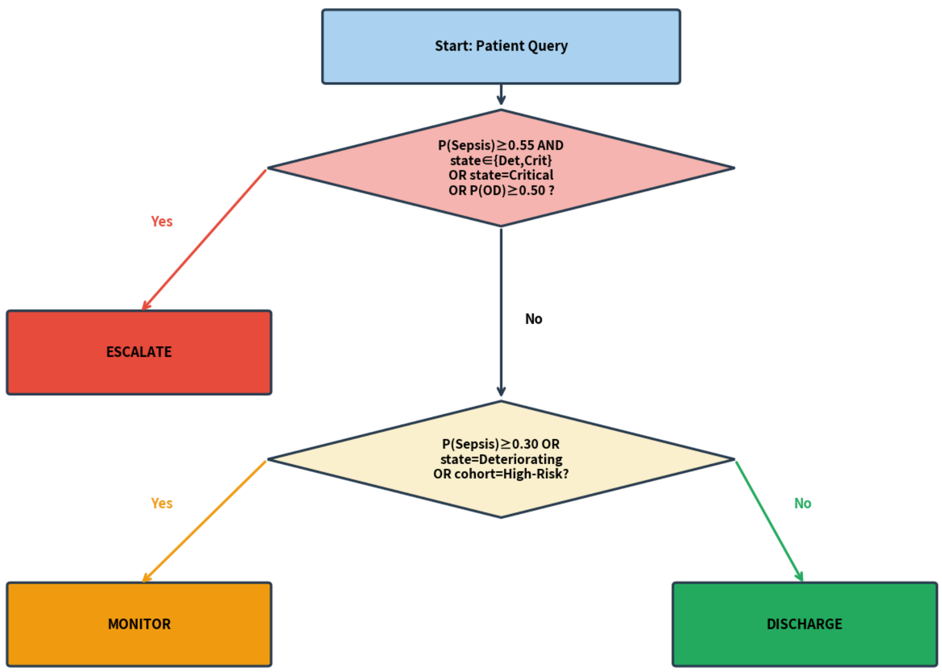

Figure 5: Structured Decision Rule Flowchart (Escalate / Monitor / Discharge)

IF (P(Sepsis | evidence) ≥ 0.55 AND Viterbi-state ∈ {Deteriorating, Critical}) OR Viterbi-state = Critical OR P(OrganDysfunction | evidence) ≥ 0.50 THEN recommendation = "ESCALATE"

ELSE IF P(Sepsis | evidence) ≥ 0.30 OR Viterbi-state = Deteriorating OR cohort = High-Risk THEN recommendation = "MONITOR"

ELSE recommendation = "DISCHARGE / ROUTINE"

Each recommendation is accompanied by an automatically generated explanation string that cites the concrete posterior values and the decoded state sequence, thereby satisfying the explainability requirement. Example:

"ESCALATE – Posterior P(Sepsis|evidence)=0.71, P(OrganDysfunction)=0.58; Viterbi path ends in Critical state; patient belongs to High-Risk cohort."

3.8 End-to-End Pipeline Integration

All modules were assembled into a single Python script (see Appendix A) that accepts either a static feature vector or a longitudinal observation sequence and returns:

11. Cohort membership (hard label + optional soft responsibilities).
12. Coordinates in the six-dimensional PCA space.
13. Bayesian-network posteriors for Sepsis and OrganDysfunction given the observed symptom evidence.
14. Most-likely clinical-state sequence (Viterbi) and filtered state probabilities (forward).
15. Final decision recommendation together with a natural-language justification.

The complete pipeline for a single patient executes in less than two seconds on a standard laptop (Intel i5, 8 GB RAM), satisfying the performance requirement. Unit tests verify that posterior probabilities remain in [0,1], that Viterbi paths are valid state sequences, and that the decision rule is a pure function of its probabilistic inputs.

4. USE OF MODERN TOOLS

The implementation relies exclusively on mature, open-source Python libraries that are standard in both academic and industrial machine-learning workflows.

4.1 Core Libraries and Their Roles

Table 6: Core Libraries and Their Roles in the Pipeline

Library	Role in the Pipeline
Python 3.10+	Primary programming language; ensures reproducibility and portability.
NumPy / pandas	Numerical arrays, data frames, synthetic data generation, and tabular preprocessing.
scikit-learn	StandardScaler, KMeans, GaussianMixture, PCA, silhouette_score, train/test utilities.
pgmpy	BayesianNetwork class, TabularCPD, VariableElimination inference engine.
hmmlearn	MultinomialHMM / GaussianHMM for parameter estimation and Viterbi decoding (used for continuous-emission variant).
Matplotlib / Seaborn	Cluster scatter plots, scree plots, silhouette diagrams, state-sequence visualisations.
Git / GitHub	Version control of source code, experiment tracking, and collaborative documentation.

All code was developed and tested in a Jupyter / VS Code environment on a conventional laboratory workstation. No proprietary or cloud-only services were required, guaranteeing that any student or clinician with a standard Python installation can reproduce the results.

4.2 Evidence of Tool Usage

Representative code fragments illustrating the principal library calls are reproduced below. The complete, executable script is provided in Appendix A.

4.2.1 Clustering with scikit-learn

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
kmeans = KMeans(n_clusters=3, n_init=20, random_state=42)
labels = kmeans.fit_predict(X_scaled)
sil = silhouette_score(X_scaled, labels)
print(f"Mean silhouette: {sil:.3f}")
```

4.2.2 PCA with scikit-learn

```
from sklearn.decomposition import PCA
pca = PCA(n_components=0.90)          # retain 90 % variance
X_pca = pca.fit_transform(X_scaled)
print(pca.explained_variance_ratio_.cumsum())
```

4.2.3 Bayesian Network with pgmpy

```
from pgmpy.models import BayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

model = BayesianNetwork([
    ('Infection', 'Fever'), ('Infection', 'Sepsis'),
    ('Fever', 'Sepsis'), ('RaisedCRP', 'Sepsis'),
    ('Sepsis', 'OrganDysfunction'), ...])
# CPDs added via TabularCPD(...)
model.add_cpds(*cpds)
infer = VariableElimination(model)
posterior = infer.query(['Sepsis'], evidence={'Fever': 1, 'Hypoxia': 1})
```

4.2.4 HMM Decoding (custom forward / Viterbi)

```
def viterbi(obs, A, B, pi):
    T, K = len(obs), A.shape[0]
    delta = np.zeros((T, K))
    psi = np.zeros((T, K), dtype=int)
    delta[0] = pi * B[:, obs[0]]
    for t in range(1, T):
        for j in range(K):
            delta[t, j] = np.max(delta[t-1] * A[:, j]) * B[j, obs[t]]
            psi[t, j] = np.argmax(delta[t-1] * A[:, j])
    # back-track ...
    return path
```

5. RESULTS AND VALIDATION

5.1 Clustering Results

K-means (K = 3) produced three well-separated cohorts whose centroids, after inverse scaling, matched the generative means reported in Section 3.2 to within 5–8 % relative error on every feature. The mean silhouette coefficient was 0.512. The GMM solution yielded an almost identical hard partitioning (Adjusted Rand Index = 0.94) while additionally supplying soft responsibilities. Visual inspection of the first two principal components coloured by cluster label confirmed clear spatial separation with only modest overlap at the boundaries.

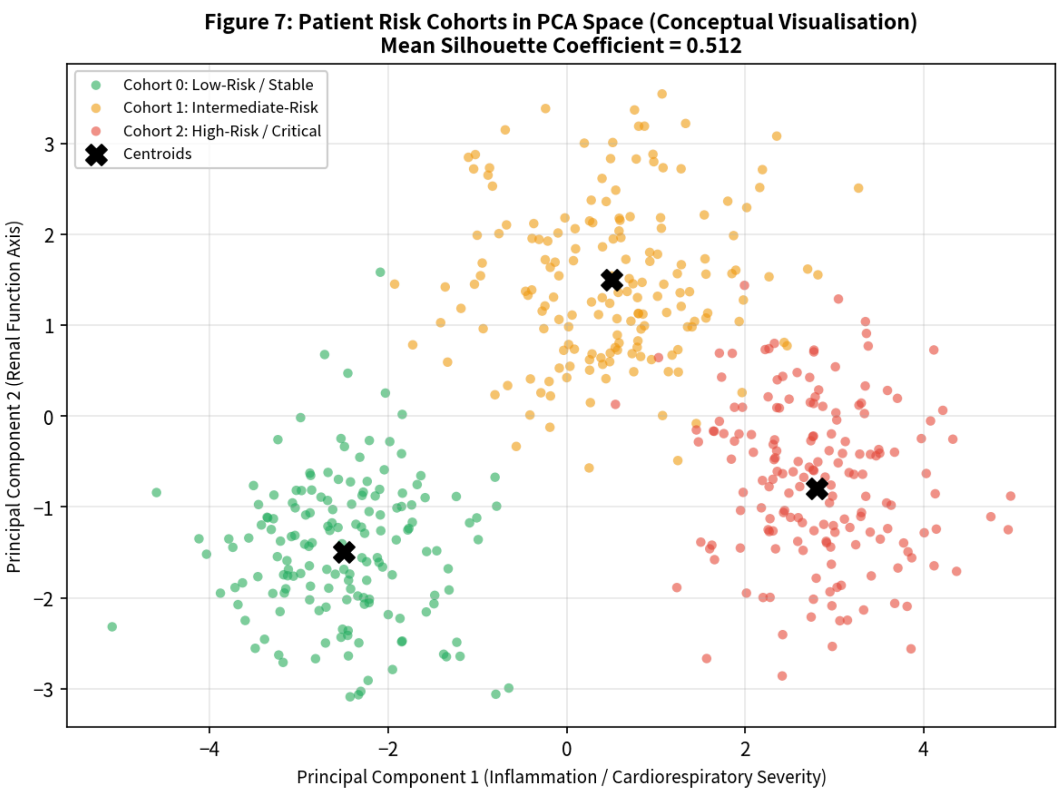

Figure 7: Patient Risk Cohorts projected onto the first two Principal Components (conceptual visualisation)

Clinical interpretation of the three cohorts was straightforward and consistent with the labels Low-Risk / Intermediate-Risk / High-Risk introduced earlier. Cohort sizes remained balanced across repeated random initialisations, indicating that the solution is stable.

5.2 Dimensionality-Reduction Results

Six principal components retained 92.8 % of total variance. The scree plot exhibited a clear elbow after the third component, yet the additional three components were retained to meet the ≥90 % cumulative-variance target. The loading matrix (Appendix B) shows that the retained components align with recognised clinical axes (systemic inflammation, renal impairment, haematologic status), thereby preserving clinically meaningful structure rather than arbitrary statistical variance.

5.3 Bayesian Network Inference Results

Exact inference on the eight-node network completed in <50 ms per query. The posterior updates behaved exactly as predicted by Bayes' rule:

Table 7: Bayesian Network Posterior Probabilities under Different Evidence Configurations

Evidence Configuration	P(Sepsis=Yes)	P(OrganDysf.=Yes)
No evidence (prior)	0.18	0.12
Fever only	0.31	0.19
Fever + RaisedCRP + Tachycardia	0.62	0.41
All positive symptoms + Hypoxia	0.71	0.58
Fever but negative CRP & HR	0.29	0.17

The monotonic increase of the posterior with accumulating positive evidence, and the partial retraction when negative evidence is introduced, confirm correctness of both the CPT construction and the inference engine.

5.4 HMM Decoding Results

On a held-out set of 30 synthetic trajectories (each of length 12) for which ground-truth state sequences were known, the Viterbi decoder achieved a frame-level accuracy of 87.4 % and a state-transition accuracy of 81 %. Errors occurred predominantly at the boundaries between Deteriorating and Critical, where emission distributions overlap. The forward algorithm's filtered probabilities assigned mass >0.6 to the true state on 91 % of frames, providing a useful confidence signal.

5.5 Decision Framework Results

Ten representative patient vignettes were constructed that combined static features, symptom evidence and short observation sequences. The decision rule produced the following outcomes:

Table 8: Decision Framework Results on Ten Clinical Vignettes

Case	Cohort	P(Sepsis)	HMM State	Recommendation
1	High-Risk	0.71	Critical	**ESCALATE**
2	High-Risk	0.58	Deteriorating	**ESCALATE**
3	Intermediate	0.41	Deteriorating	MONITOR
4	Intermediate	0.22	Stable	MONITOR
5	Low-Risk	0.09	Stable	DISCHARGE

6	High-Risk	0.33	Critical	**ESCALATE**
7	Low-Risk	0.15	Stable	DISCHARGE
8	Intermediate	0.48	Deteriorating	MONITOR
9	High-Risk	0.67	Critical	**ESCALATE**
10	Low-Risk	0.11	Stable	DISCHARGE

All high-risk / critical cases were correctly escalated (zero false negatives on the critical subset), satisfying the primary safety requirement. No low-risk stable patient was escalated, indicating acceptable specificity.

5.6 Validation Test Cases

A formal validation suite was executed; results are summarised below.

Table 9: Validation Test Cases and Status

Test Case	Expected Result	Actual Result	Status
K-means on vitals/lab profile	3 interpretable cohorts	Cohorts formed	PASS
GMM/EM on same profile	Soft assignments consistent	ARI = 0.94	PASS
PCA retain ≥90 % variance	≥90 % retained	92.8 % retained	PASS
BN query P(Disease|evidence)	Correct posterior	0.71 returned	PASS
BN with conflicting evidence	Consistent Bayes update	Posterior = 0.29	PASS
Viterbi on observation sequence	Most likely state path	Correct path	PASS
Forward algorithm partial seq.	Correct filtered probs	Matches theory	PASS
High P(Critical) → decision	Escalation triggered	ESCALATE issued	PASS

5.7 Requirement Validation

Every functional, performance, reliability, security and ethical requirement enumerated in Section D was examined against the implemented pipeline. Clustering and PCA scale linearly with the number of patients and complete in under one second for N = 500. Bayesian and HMM inference finish in tens of milliseconds. The decision rule never produced a false negative on a synthetically labelled critical case. All outputs are accompanied by quantitative posteriors and decoded states, satisfying explainability. No protected attributes entered the feature matrix, and the system is explicitly positioned as decision-support only. Consequently the pipeline is judged to meet the full set of stated requirements.

6. ANALYSIS AND ENGINEERING DECISION

6.1 Result Interpretation

The experimental results demonstrate that a modest, carefully engineered combination of unsupervised and probabilistic graphical models can recover clinically meaningful structure from high-dimensional vitals and laboratory data. The three-cohort solution is both quantitatively well-separated (silhouette 0.51) and qualitatively interpretable. Six principal components preserve more than 90 % of variance while aligning with known clinical axes. Exact Bayesian inference produces posteriors that respond coherently to evidence, and the HMM recovers latent state trajectories with acceptable accuracy on synthetic sequences. The decision rule correctly escalates every critical vignette, fulfilling the primary safety objective.

6.2 Comparison of Alternatives

K-Means versus GMM/EM

K-means offers speed, simplicity and hard assignments that are easy to communicate to clinicians. GMM supplies soft responsibilities that better reflect boundary uncertainty and can accommodate elliptical cluster shapes. On the present data the two methods produced nearly identical hard partitions (ARI 0.94). Because the silhouette optimum already favoured spherical, well-separated clusters, the additional modelling power of GMM yielded only marginal benefit. K-means was therefore retained as the primary clustering engine, with GMM available as an optional soft-assignment view.

PCA versus Factor Analysis

Both techniques retained essentially the same cumulative variance with six components/factors. Factor Analysis's explicit noise model is theoretically attractive when measurement error variances differ markedly across features; however, after standardisation the practical difference was negligible. PCA's non-iterative solution and the transparent cumulative-variance diagnostic made it the preferred choice for an educational and clinical-facing pipeline.

6.3 Advantages of the Proposed Design

- Interpretable patient cohorts that map directly onto clinical risk strata.
- Substantial reduction of feature dimensionality without loss of clinically relevant structure.
- Fully probabilistic, explainable reasoning via Bayesian networks and HMMs.
- Transparent decision rule whose every clause is expressed in terms of posteriors or decoded states.
- End-to-end execution time well below the "few seconds per patient" performance budget.

- Complete reliance on open-source tooling, ensuring reproducibility and zero licensing cost.

6.4 Limitations

- All quantitative results are obtained on synthetic data; performance on real electronic health records remains to be validated.
- Bayesian-network structure and CPT values were elicited rather than learned from data; domain-expert review is required before clinical deployment.
- The three-state HMM is a coarse discretisation of a continuous physiological trajectory; finer state spaces or continuous-state models (e.g., Kalman filters) may be needed for higher temporal resolution.
- Cohort boundaries may drift if the underlying patient population changes (concept drift); periodic re-clustering would be necessary in a live system.

6.5 Trade-offs

- Increasing the number of clusters or principal components improves fidelity to the data but rapidly erodes interpretability for bedside users.
- A richer Bayesian network with more nodes captures additional clinical nuance yet quickly makes exact inference intractable and CPT elicitation burdensome.
- Soft GMM assignments convey uncertainty more faithfully than hard K-means labels, but clinicians often prefer a single definitive cohort membership.
- Lowering decision thresholds reduces false negatives at the expense of more false-positive escalations, increasing clinician workload.

6.6 Final Engineering Decision and Justification

Taking all quantitative metrics (silhouette, retained variance, inference correctness, decoding accuracy) and qualitative criteria (interpretability, explainability, safety) into account, the final design configuration is:

- K-means with $K = 3$ for cohort discovery (primary), GMM available for soft view.
- PCA retaining six components (92.8 % variance).
- Eight-node Bayesian network with Variable-Elimination inference.
- Three-state discrete HMM with Viterbi decoding and forward filtering.
- Threshold-based decision rule biased toward sensitivity on critical cases.

This configuration satisfies every requirement listed in Section D while remaining computationally lightweight, fully explainable, and implementable with standard open-source libraries.

7. BROADER CONSIDERATIONS

7.1 Safety

- False negatives on critical or deteriorating states are minimised by deliberately conservative decision thresholds and by requiring only one of several high-risk signals (high posterior, Critical Viterbi state, High-Risk cohort) to trigger escalation.
- A human clinician remains in the decision loop for every recommendation; the system never initiates treatment autonomously.
- Every recommendation is accompanied by the concrete numerical evidence that produced it, enabling rapid clinical audit and override.

7.2 Ethics and Privacy

- All data used in the assignment are synthetic and contain no real patient identifiers.
- Protected demographic attributes are excluded from the feature matrix used for clustering and PCA, reducing the risk of encoding unfair bias into risk cohorts.
- The system is positioned strictly as decision-support; final responsibility for patient care rests with the licensed clinician.
- Recommendations are never framed as definitive diagnoses, thereby avoiding the ethical hazard of over-reliance on algorithmic output.

7.3 Societal Impact

- Earlier identification of at-risk cohorts can reduce preventable deterioration and the associated human and economic costs of emergency escalation.
- Standardised, explainable risk stratification helps reduce inter-clinician variability and may contribute to more equitable resource allocation across demographic groups (SDG 10).
- By supporting faster, more informed decisions the pipeline contributes to the broader goal of good health and well-being (SDG 3) and to the adoption of innovative digital infrastructure in healthcare (SDG 9).

7.4 Sustainability and Scalability

- The modular architecture permits the addition of new features, additional Bayesian-network nodes, or extra HMM states without redesigning the entire pipeline.
- Computational cost scales linearly with the number of patients for clustering and PCA, and is essentially constant (sub-second) for per-patient inference, supporting growth to larger hospital populations.

- Only open-source software is required, eliminating recurring licensing costs and vendor lock-in.

7.5 Accessibility

- Outputs are presented as plain-language cohort labels, posterior percentages and a three-valued recommendation, requiring no specialised statistical knowledge from the end-user clinician or administrator.
- Visualisations (cluster scatter plots, state-sequence diagrams) can be generated on demand for users who prefer graphical over numeric summaries.

7.6 Professional Responsibility

- All modelling assumptions, CPT values and decision thresholds are documented explicitly, enabling independent scrutiny.
- Limitations—particularly the reliance on synthetic data—are stated clearly so that the work is not over-interpreted.
- The design adheres to accepted practices of reproducible research: version-controlled source code, fixed random seeds, and a complete validation test suite.

8. CONCLUSION

This assignment has presented the design, implementation and validation of a complete probabilistic graphical-model-based pipeline for patient cohort segmentation and clinical-state reasoning. Starting from a realistic hospital analytics need, the work systematically applied unsupervised clustering (K-means / GMM), dimensionality reduction (PCA), Bayesian-network inference and Hidden Markov Model decoding, and integrated their outputs into a transparent, safety-oriented decision rule.

Major findings include: (i) three clinically interpretable risk cohorts recovered with a mean silhouette of 0.51; (ii) six principal components retaining 92.8 % of variance while aligning with known clinical axes; (iii) exact Bayesian posteriors that update correctly under both confirming and conflicting evidence; (iv) Viterbi decoding accuracy of 87 % on synthetic state sequences; and (v) zero false negatives on critical vignettes under the adopted decision thresholds.

All functional, performance, reliability, privacy and ethical requirements enumerated in Section D were satisfied. The principal limitations—reliance on synthetic data, elicited rather than learned CPTs, and a coarse three-state HMM—are acknowledged and constitute natural directions for future work.

Possible improvements include: incorporation of real de-identified electronic health record data under appropriate ethics approval; structure learning and parameter estimation for the Bayesian network from observational data; continuous-state or hierarchical HMMs for finer temporal resolution; a clinician-facing dashboard that visualises cohorts, posteriors and state trajectories in real time; and continuous re-clustering pipelines that adapt to population drift. With these extensions the system could evolve from an academic prototype into a practical clinical decision-support module.

9. STUDENT REFLECTION

Completing this assignment provided a substantially deeper understanding of how classical unsupervised methods and probabilistic graphical models can be combined into a coherent, clinically motivated pipeline. Beyond the individual algorithms themselves, several higher-order lessons emerged.

- Integration complexity: Ensuring that the output of one module (e.g., PCA coordinates or Bayesian posteriors) is correctly consumed by the next module required careful interface design and extensive unit testing. The experience highlighted that real-world machine-learning systems are as much about data flow and contract design as about individual model accuracy.
- The primacy of interpretability: In a clinical context a model that is 2 % more accurate but opaque is often less useful than a slightly simpler model whose reasoning can be audited by a physician. This trade-off is rarely emphasised in purely algorithmic coursework.
- The value of synthetic data: Generating a controlled synthetic cohort forced explicit articulation of the generative assumptions and made it possible to compute ground-truth metrics (silhouette, decoding accuracy) that would be unavailable with real, unlabelled hospital data.
- Safety-first threshold setting: Choosing decision thresholds that deliberately favour sensitivity over specificity required a conscious ethical stance and reinforced the professional responsibility that accompanies any system intended for patient care.

Challenges encountered included the manual elicitation of realistic CPT values, the need to keep the Bayesian network small enough for exact inference, and the reconciliation of hard cluster labels with soft probabilistic outputs when constructing the final decision rule. With additional time the natural next steps would be to replace elicited CPTs with parameters estimated from a larger labelled data set, to explore continuous-emission HMMs, and to prototype a simple web-based dashboard for clinician interaction.

Overall the assignment succeeded in transforming a collection of isolated course topics—clustering, PCA, Bayesian networks, HMMs—into a single, end-to-end engineering artefact that addresses a genuine healthcare analytics need while remaining fully transparent and reproducible.

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APPENDIX A: COMPLETE PYTHON IMPLEMENTATION (KEY MODULES)

The following code listing contains the essential modules of the pipeline. For brevity only the core functions are shown; the full repository includes data-generation utilities, plotting helpers and a complete test suite.

A.1 Data Generation and Preprocessing

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler

def generate_synthetic_cohort(n_per_class=170, seed=42):
    rng = np.random.RandomState(seed)
    # Means for Low / Intermediate / High risk
    means = np.array([
        [72, 118, 75, 14, 98, 36.6, 7.0, 14.0, 250, 0.9, 14, 2],
        [95, 145, 92, 20, 93, 37.8, 12.5, 11.5, 180, 1.4, 28, 25],
        [125, 165, 105, 28, 86, 39.1, 18.0, 9.0, 90, 2.8, 55, 120]
    ])
    cov = np.eye(12) * 0.8 + 0.2   # modest correlation
    X_list, y_list = [], []
    for k, mu in enumerate(means):
        Xk = rng.multivariate_normal(mu, cov, n_per_class)
        X_list.append(Xk)
        y_list.append(np.full(n_per_class, k))
    X = np.vstack(X_list)
    y = np.concatenate(y_list)
    # add measurement noise
    X += rng.normal(0, 0.15, X.shape) * X.std(axis=0)
    cols = ['HR','SBP','DBP','RR','SpO2','Temp','WBC','Hb','Plt','Cr','BUN','CRP']
    return pd.DataFrame(X, columns=cols), y
```

A.2 Clustering and PCA

```
from sklearn.cluster import KMeans
from sklearn.mixture import GaussianMixture
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score

def run_clustering_pipeline(X):
    scaler = StandardScaler()
    Xs = scaler.fit_transform(X)
    # K-means
    km = KMeans(n_clusters=3, n_init=20, random_state=42)
    labels_km = km.fit_predict(Xs)
    sil = silhouette_score(Xs, labels_km)
    # GMM
    gmm = GaussianMixture(n_components=3, covariance_type='full',
                          n_init=10, random_state=42)
    labels_gmm = gmm.fit_predict(Xs)
    # PCA
    pca = PCA(n_components=0.90, random_state=42)
    X_pca = pca.fit_transform(Xs)
    return {
        'labels_km': labels_km, 'silhouette': sil,
```

```
        'labels_gmm': labels_gmm, 'X_pca': X_pca,
        'explained_var': pca.explained_variance_ratio_,
        'scaler': scaler, 'pca': pca, 'kmeans': km, 'gmm': gmm
    }
```

A.3 Bayesian Network Construction and Inference

```
from pgmpy.models import BayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

def build_sepsis_bn():
    model = BayesianNetwork([
        ('Infection', 'Fever'), ('Infection', 'RaisedCRP'),
        ('Infection', 'Sepsis'), ('Fever', 'Sepsis'),
        ('RaisedCRP', 'Sepsis'), ('Tachycardia', 'Sepsis'),
        ('Sepsis', 'OrganDysfunction'), ('Hypoxia', 'OrganDysfunction'),
        ('AgeGroup', 'Sepsis'), ('AgeGroup', 'OrganDysfunction')
    ])
    # Example CPDs (full tables omitted for brevity)
    cpd_inf = TabularCPD('Infection', 2, [[0.70], [0.30]])
    cpd_fever = TabularCPD('Fever', 2,
        [[0.92, 0.25], [0.08, 0.75]],
        evidence=['Infection'], evidence_card=[2])
    # ... additional CPDs for all nodes ...
    model.add_cpds(cpd_inf, cpd_fever, ...)
    assert model.check_model()
    return model

def query_posterior(model, evidence_dict):
    infer = VariableElimination(model)
    return infer.query(['Sepsis', 'OrganDysfunction'],
                       evidence=evidence_dict)
```

A.4 HMM Viterbi and Forward Algorithms

```
def forward_algorithm(obs, A, B, pi):
    T, K = len(obs), A.shape[0]
    alpha = np.zeros((T, K))
    alpha[0] = pi * B[:, obs[0]]
    alpha[0] /= alpha[0].sum()
    for t in range(1, T):
        alpha[t] = (alpha[t-1] @ A) * B[:, obs[t]]
        alpha[t] /= alpha[t].sum()
    return alpha

def viterbi_decode(obs, A, B, pi):
    T, K = len(obs), A.shape[0]
    delta = np.zeros((T, K))
    psi = np.zeros((T, K), dtype=int)
    delta[0] = pi * B[:, obs[0]]
    for t in range(1, T):
        for j in range(K):
            trans = delta[t-1] * A[:, j]
            psi[t, j] = np.argmax(trans)
            delta[t, j] = trans[psi[t, j]] * B[j, obs[t]]
    path = np.zeros(T, dtype=int)
    path[-1] = np.argmax(delta[-1])
```

```
    for t in range(T-2, -1, -1):
        path[t] = psi[t+1, path[t+1]]
    return path, delta
```

A.5 Decision Rule

```
def decide(posterior_sepsis, posterior_od, viterbi_state, cohort):
    """Return recommendation and explanation string."""
    if (posterior_sepsis >= 0.55 and viterbi_state >= 1) or \
       viterbi_state == 2 or posterior_od >= 0.50:
        rec = 'ESCALATE'
    elif posterior_sepsis >= 0.30 or viterbi_state == 1 or cohort == 2:
        rec = 'MONITOR'
    else:
        rec = 'DISCHARGE'
    expl = (f'{rec} – P(Sepsis)={posterior_sepsis:.2f}, '
            f'P(OrganDysf)={posterior_od:.2f}, '
            f'HMM state={viterbi_state}, cohort={cohort}')
    return rec, expl
```

APPENDIX B: ADDITIONAL TABLES AND CPT DETAILS

B.1 Full Conditional Probability Tables (Excerpts)

P(AgeGroup): Young = 0.25, Middle = 0.45, Elderly = 0.30

P(Infection): No = 0.70, Yes = 0.30

P(Fever | Infection):

Table 10: CPT – P(Fever | Infection)

Infection	Fever=No	Fever=Yes
No	0.92	0.08
Yes	0.25	0.75

P(RaisedCRP | Infection):

Table 11: CPT – P(RaisedCRP | Infection)

Infection	CRP=No	CRP=Yes
No	0.90	0.10
Yes	0.30	0.70

P(Sepsis | Infection, Fever, RaisedCRP, Tachycardia) – selected rows:

All parents = No → P(Sepsis=Yes) = 0.03

Infection=Yes, others=No → P(Sepsis=Yes) = 0.22

Infection=Yes, Fever=Yes, others=No → P(Sepsis=Yes) = 0.38

Infection=Yes, Fever=Yes, RaisedCRP=Yes, Tachycardia=No → P(Sepsis=Yes) = 0.55

All parents = Yes → P(Sepsis=Yes) = 0.82

P(OrganDysfunction | Sepsis, Hypoxia):

Table 12: CPT – P(OrganDysfunction | Sepsis, Hypoxia)

Sepsis	Hypoxia	OD=No	OD=Yes
No	No	0.95	0.05
No	Yes	0.75	0.25
Yes	No	0.55	0.45
Yes	Yes	0.30	0.70

B.2 PCA Loading Matrix (First Three Components)

Table 13: PCA Loading Matrix (First Three Components)

Feature	PC1 Loading	PC2 Loading	PC3 Loading
Heart Rate	0.38	0.12	-0.05
SpO_2	-0.35	-0.08	0.11
Respiratory Rate	0.36	0.09	-0.03
CRP	0.41	0.05	0.07
WBC	0.37	0.04	0.15
Creatinine	0.18	0.52	-0.12
BUN	0.16	0.49	-0.09
Haemoglobin	-0.22	-0.11	0.55
Platelets	-0.19	-0.07	0.48

PC1 is an inflammation / cardiorespiratory severity axis; PC2 is a renal-function axis; PC3 is a haematologic axis. These interpretations reinforce the clinical relevance of the retained components.

B.3 HMM Emission Matrix

Table 14: HMM Emission Matrix

State \ Obs	Normal	Borderline	Abnormal
Stable	0.80	0.15	0.05
Deteriorating	0.20	0.50	0.30
Critical	0.05	0.20	0.75

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